

Global YouTube Statistics



Final Data Analytics Case Study —
SWYNEX Task 4



Project Overview

This project analyzes global YouTube channel statistics to explore subscriber performance, video views, category distribution, geographical representation, and upload activity.

The objective is to transform raw channel data into meaningful insights using data cleaning, exploratory data analysis, and interactive dashboard visualization.



Problem Statement

YouTube channels differ in audience size, content category, geographical presence, and upload frequency. Understanding these differences can help content creators and channel managers compare performance and identify patterns.

This project investigates which channels lead in subscriber counts, which categories generate the most views, where channels are concentrated, and whether higher upload volume is associated with stronger channel performance.



Dataset Information

- Dataset name: Global YouTube Statistics 2023
- Source: "C:\Users\shado\Downloads\global_youtube_statistics_2023.csv"
- Original records: 995
- Final records: 554
- Number of columns: 28
- Key variables: Channel name, country, category, subscribers, video views, and uploads, subject to the columns available in the dataset.

Data Cleaning and Preparation



The implemented cleaning workflow included:

1. Loading the CSV dataset.
2. Inspecting the first ten rows, dataset dimensions, column names, and data types.
3. Checking missing values and duplicate records.
4. Standardizing column names to lowercase and replacing spaces with underscores.
5. Removing duplicate rows.
6. Trimming leading and trailing spaces from text fields.
7. Reviewing numerical descriptive statistics.
8. Checking missing values and duplicate records after cleaning.
9. Exporting the cleaned dataset as "cleaned_dataset.csv".
10. Generating a data-cleaning report.

Cleaning results: "C:\Users\shado\swynex ass1.ipynb"

Exploratory Data Analysis



The EDA workflow uses Pandas, NumPy, Matplotlib, and Seaborn.

The analysis includes:

- Dataset structure and data types.
- Mean, median, standard deviation, minimum, and maximum
- Top ten channels by subscribers.
- Category distribution.
- Subscriber distribution.
- Numerical correlation analysis.
- IQR-based outlier detection.

The generated outputs include charts, summary statistics, a correlation matrix, and an outlier summary.

EDA findings: "C:\Users\shado\swynex_ass2.ipynb"



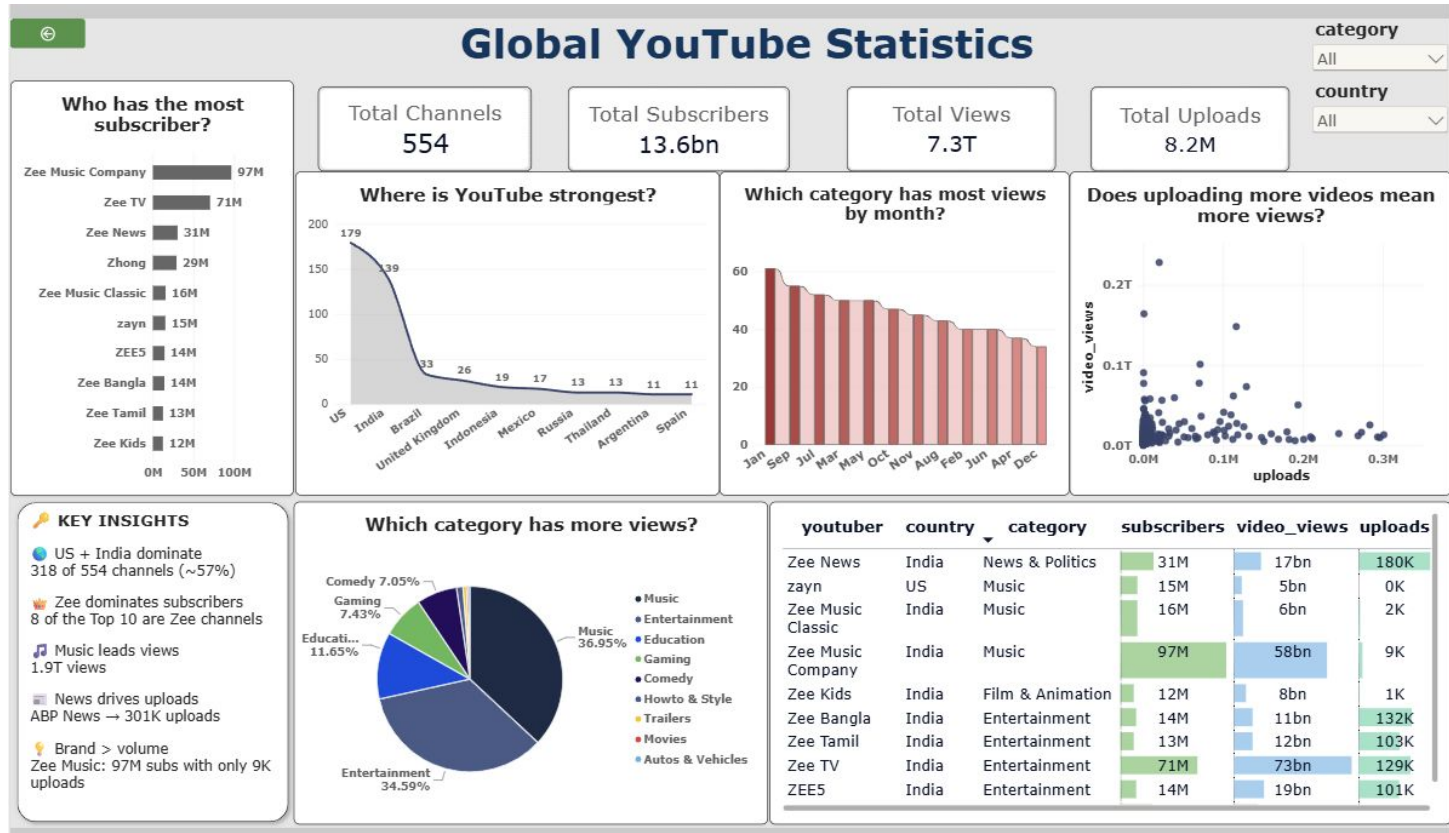
Power BI Dashboard

The Power BI dashboard presents the Global YouTube Statistics 2023 dataset through KPI cards, category and country charts, channel rankings, a scatter plot, and a channel-level data table.

The dashboard displays these headline metrics:

- Total channels: 554
- Total subscribers: 13.6 billion
- Total views: 7.3 trillion
- Total uploads: 8.2 million

The dashboard allows users to explore country and category filters, compare leading channels, and investigate differences in views and upload activity.





Key Business Insights

- 1. Subscriber leadership:** Zee Music Company is displayed with 97 million subscribers.
- 2. Geographical representation:** The dashboard highlights the US and India, with 318 channels combined under the displayed metric.
- 3. Category performance:** Music accounts for 36.95% of category views in the dashboard.
- 4. Upload activity:** ABP News is displayed with approximately 301,000 uploads.
- 5. Upload volume versus views:** The scatter plot shows variation in views across upload levels. A statistical correlation calculation is needed before concluding the strength or direction of the relationship.



Business Recommendations

- Compare channels using subscribers, views, and upload activity together.
- Investigate category-specific performance before planning content.
- Compare channel performance across countries while accounting for differences in dataset coverage.
- Evaluate upload efficiency using views per upload, where the required data is reliable.
- Avoid treating correlation as proof that upload frequency causes higher views.



Limitations

The findings depend on the dataset's coverage, field definitions, missing values, and reporting period. Extreme values may influence aggregate metrics. The analysis is observational and does not establish causal relationships.



Conclusion

This project demonstrates a complete data analytics workflow, from CSV inspection and data preparation to exploratory analysis, visualization, and business recommendations. The resulting dashboard brings key channel metrics together to support clearer comparisons and data-informed decisions.

Github link : https://github.com/hilmiyapsana87-code/Swynex_ass1.git

